

Overview of Gaussian process

Subject: Gaussian process

$$f(x) \sim \mathcal{GP}(m(x), k(x, x'))$$

1.

$$\mathbb{E}[X_t X_{t+h}] = \sum_{n=1}^{\infty} c_n^2 \cos(\lambda_n h) \quad \{\displaystyle \mathbb{E}[X_t X_{t+h}] = \sum_{n=1}^{\infty} c_n^2 \cos \lambda_n h\}$$

2.

$$K_{\text{Matérn}}(x, x') = \frac{2^{1-\nu} \Gamma(\nu)}{(2\nu d)^{\nu}} K_{\nu} \left(\frac{\sqrt{2\nu} d}{\ell} \right) \quad \{\displaystyle K_{\text{Matérn}}(x, x') = \frac{2^{1-\nu} \Gamma(\nu)}{(2\nu d)^{\nu}} K_{\nu} \left(\frac{\sqrt{2\nu} d}{\ell} \right)\}$$

3.

$$X_t = \sum_{n=1}^{\infty} c_n (\xi_n \cos(\lambda_n t) + \eta_n \sin(\lambda_n t)), \quad \{\displaystyle X_t = \sum_{n=1}^{\infty} c_n (\xi_n \cos \lambda_n t + \eta_n \sin \lambda_n t)\}$$

4.

$$\sigma(h) = 2 \mathbb{E}[X_t X_t] - 2 \mathbb{E}[X_t X_{t+h}] = 2 \sum_{n=1}^{\infty} c_n^2 \sin^2(\lambda_n h) \quad \{\displaystyle \sigma(h) = \sqrt{2 \mathbb{E}[X_t X_t] - 2 \mathbb{E}[X_t X_{t+h}]} = 2 \sqrt{\sum_{n=1}^{\infty} c_n^2 \sin^2 \left(\frac{\lambda_n h}{2} \right)}}\}$$

5.

When concerned with a general Gaussian process regression problem (Kriging), it is assumed that for a Gaussian process f observed at coordinates x , the vector of values $f(x)$ is just one sample from a multivariate Gaussian distribution of dimension equal to the number of observed coordinates n . Therefore, under the assumption of a zero-mean distribution, $f(x') \sim N(0, K(\theta, x, x'))$, where $K(\theta, x, x')$ is the covariance matrix between all possible pairs (x, x') for a given set of hyperparameters θ . As such the log marginal likelihood is:

6.

Some history. Sufficiency was announced by Xavier Fernique in 1964, but the first proof was published by Richard M. Dudley in 1967. Necessity was proved by Michael B. Marcus and Lawrence Shepp in 1970. There exist sample continuous processes X such that $I(\sigma) = \infty$; they violate condition (*). An example found by Marcus and Shepp is a random lacunary Fourier series

7.

As such, almost all sample paths of a mean-zero Gaussian process with positive definite kernel K will lie outside of the Hilbert space $H(K)$.

8.

Stationarity For general stochastic processes strict-sense stationarity implies wide-sense stationarity but not every wide-sense stationary stochastic process is strict-sense stationary. However, for a Gaussian stochastic process the two concepts are equivalent. A Gaussian stochastic process is strict-sense stationary if and only if it is wide-sense stationary.

9.

Physical applications Gaussian processes have found increasing applications in many domains of the natural sciences due to their statistical modelling properties. Molecular property prediction has employed these process models in small molecular datasets due to their inference capabilities and computational costs. They are also being increasingly used as surrogate models for force field optimization.

10.

A key fact of Gaussian processes is that they can be completely defined by their second-order statistics. Thus, if a Gaussian process is assumed to have mean zero, defining the covariance function completely defines the process' behaviour. Importantly the non-negative definiteness of this function enables its spectral decomposition using the Karhunen–Loève expansion. Basic aspects that can be defined through the covariance function are the process' stationarity, isotropy, smoothness and periodicity. Stationarity refers to the process' behaviour regarding the separation of any two points x and x' . If the process is stationary, the covariance function depends only on $x - x'$. For example, the Ornstein–Uhlenbeck process is stationary. If the process depends only on $|x - x'|$, the Euclidean distance (not the direction) between x and x' , then the process is considered isotropic. A process that is concurrently stationary and isotropic is considered to be homogeneous; in practice these properties reflect the differences (or rather the lack of them) in the behaviour of the process given the location of the observer. Ultimately Gaussian processes translate as taking priors on functions and the smoothness of these priors can be induced by the covariance function. If we expect that for "near-by" input points x and x' their corresponding output points y and y' to be "near-by" also, then the assumption of continuity is present. If we wish to allow for significant displacement then we might choose a rougher covariance function. Extreme examples of the behaviour is the Ornstein–Uhlenbeck covariance function and the squared exponential where the former is never differentiable and the latter infinitely differentiable. Periodicity refers to inducing periodic patterns within the behaviour of the process. Formally, this is achieved by mapping the input x to a two dimensional vector $u(x) = (\cos(x), \sin(x))$.

11.

$$\mathbb{E} \sum_{n \in \mathbb{N}} (|\xi_n| + |\eta_n|) = \sum_{n \in \mathbb{N}} \mathbb{E} [|\xi_n| + |\eta_n|] = \text{const} \cdot \sum_{n \in \mathbb{N}} c_n < \infty, \quad \{\text{textstyle } \mathbb{E}\}$$

$$\sum_{n \in \mathbb{N}} c_n (|\xi_n| + |\eta_n|) = \sum_{n \in \mathbb{N}} c_n \mathbb{E} [|\xi_n| + |\eta_n|] = \sum_{n \in \mathbb{N}} c_n \mathbb{E} [|\xi_n| + |\eta_n|] = \sum_{n \in \mathbb{N}} c_n \mathbb{E} [|\xi_n| + |\eta_n|]$$

$$|c_n| \leq \text{const} \cdot \sum_{n=1}^{\infty} c_n < \infty,$$

12.

whence $\sum_{n=1}^{\infty} c_n (|\xi_n| + |\eta_n|) < \infty$ almost surely, which ensures uniform convergence of the Fourier series almost surely, and sample continuity of X .

13.

In practical applications, Gaussian process models are often evaluated on a grid leading to multivariate normal distributions. Using these models for prediction or parameter estimation using maximum likelihood requires evaluating a multivariate Gaussian density, which involves calculating the determinant and the inverse of the covariance matrix. Both of these operations have cubic computational complexity which means that even for grids of modest sizes, both operations can have a prohibitive computational cost. This drawback led to the development of multiple approximation methods.